Roll No. ....

TCE–301

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

MECHANICS OF FLUIDS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the different types of hydraulic similarities that must exist between a prototype and its model. (CO1)

(b) A rectangular solid block of 1 m by 1 m that weighs 30 N slides down a 30° inclined plane. The plane is lubricated by a 5 mm thick film of oil of viscosity of 0.04 N-s/m^2. Find the terminal velocity of the block. (CO1)

(c) In the model test of a spillway the discharge and velocity of flow over the model were 2 m^3/s and 1.5 m/s respectively. Calculate the velocity and discharge over the prototype which is 36 times the model size.

(CO1)

P. T. O.

2. (a) Derive an expression for the forces exerted on a submerged vertical plane surface by the static liquid and locate the position of centre of pressure. (CO2)

   (b) A circular plate of diameter 3 m is submerged in water vertically such that its centre is 2 m below the free surface of the water. Determine the total pressure force on the plate and the position of the centre of pressure. (CO2)

   (c) The velocity potential function is given by $\Phi = 5\,(x^2 - y^2)$. Calculate the velocity components at the point (4, 5). (CO2)

3. (a) Derive the expression for time of emptying a tank through an orifice at its bottom. (CO3)

   (b) A pipe 300 m long has a slope of 1 in 100 and tapers from 1 m diameter at the higher end to 0.5 m at the lower end. The quantity of water flowing is 900 liters/second. If the pressure at the higher end is 70 kPa, find the pressure at the lower end. (CO3)

   (c) A rectangular channel 2 m wide has a discharge of 250 lit./sec, which is measured by a right angled V-Notch. Find the position of the apex of the notch from the bed of the channel if maximum depth of water is not to exceed 1.30 m. Take $C_d = 0.2$. (CO3)

4. (a) For the Laminar flow through a circular pipe, prove that the shear stress variation across the section of pipe is linear. (CO4)

   (b) Derive the relation for laminar flow through porous media i.e., Darcy's law. (CO4)

(c) Determine the pressure gradient, the shear stress at the two horizontal parallel plates and the discharge per metre width for the laminar flow of oil with a maximum velocity of 2 m/s between two horizontal parallel fixed plates which are 10 mm apart. Given $\mu = 2.4525$ Nsm^2. (CO4)

5. (a) What do you mean by surge tank ? Write down the types of a surge tank. (CO5)

(b) A main pipeline carrying water at the rate of 2.5 m^3/s has to be divided into two pipes of diameters .9 m and 0.1 m and then joined back after a length of 1800 m. The friction factor for both pipes is 0.018. Calculate the flow in each line. (CO5)

(c) A valve is provided at the end of a cast iron pipe of diameter 150 mm and of thickness 10 mm. The water is flowing through the pipe, which is suddenly stopped by closing the valve. Find the maximum velocity of water, when the rise in pressure due to sudden closure of valve is 196.2 N/cm^2. Take k for water as 19.62×104 N/cm^2 and E for cast iron pipe as 11.772×10^6 N/cm^2. (CO5)

Roll No. ....

TCE–302

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

BASIC SURVEYING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the following terms related to chain surveying with the help of a neat diagram : (CO1)

   (i) Line ranger

   (ii) Cross staff

   (iii) Optical square

   (iv) Clinometer

   (v) Engineer's chain.

   (b) Write a brief note on various methods of linear measurement. A 20 m chain used for a survey was found to be 20.10 m at the beginning and 20.30 m at the end of the work. The area of the plan drawn to a scale of 1 cm = 8 m was measured with the help of a planimeter and was found to be 32.56 sq. cm. Find the true area of the field. (CO1)

P. T. O.

(c) A tape of 30 m length suspended in catenary measured the length of a base line. After applying all corrections, the deduced length of the base line was 1462.36 m. Later on its was found that the actual pull applied was 155 N and not the 165 N as recorded in the field book. Correct the deduced length for the incorrect pull. The type was standardized on the flat under a pull of 85 N having a mass of 0.024 kg/m and cross-sectional area of 4.12 mm^2. The Young's modulus of the tape material is 152000 MN/m^2 and the acceleration due to gravity is 9.806 m/s^2. (CO1)

2. (a) A tape of 30 m length suspended in catenary measured the length of a base line. After applying all corrections, the deduced length of the base line was 1462.36 m. Later on it was found that the actual pull applied was 155 N and not the 165 N as recorded in the field book. Correct the deduced length for the incorrect pull. The type was standardized on the flat under a pull of 85 N having a mass of 0.024 kg/m and cross-sectional area of 4.12 mm^2. The Young's modulus of the tape material is 152000 MN/m^2 and the acceleration due to gravity is 9.806 m/s^2. (CO2)

(b) The following bearings were observed for a closed Traverse ABCDEA :

Line	**F. B.**
AB	140°3'
BC	80°4'
CD	340°0'
DE	290°30'
EA	230°30'

Compute all the interior angles and draw the traverse. (CO2)

(c) The fore bearings and back bearings of the lines of a closed traverse ABCDA were recorded as below : (CO2)

Line	Fore bearing	Back bearing
AB	77°30'	259°10'
BC	110°30'	289°30'
CD	228°00'	48°00'
DA	309°50'	129°10'

Determine which of the stations are affected by local attraction and compute the values of the correct bearing.

3. (a) What is meant by error of closure and balancing of a traverse ? Enlist various method of balancing a traverse and only explain Bowditch Rule. (CO3)

(b) The following data were collected while running a closed traverse ABCDA. Calculate the omitted values and draw the traverse. (CO3)

Side	Length (m)	Bearing
AB	200.0	Omitted
BC	98.0	178°
CD	Omitted	270°
DA	86.4	1°

(c) Calculate latitudes, departure and closing error for the following traverse by Transit rule. (CO3)

Line	Length	Whole Circle Bearing
AB	89.31	45°10'
BC	219.76	75°5'
CD	151.18	161°52'
DE	159.10	228°43'
EA	232.26	300°42'

4. (a) Explain the following terms related to leveling : (CO4)

(i) GTS benchmarks

(ii) Reciprocal leveling

(iii) Differential leveling

(iv) Effect of curvature of earth on leveling

(v) Tilting level

(b) The readings given in table below, were recorded in a leveling operation from points 1 to 10. Reduce the levels by the height of instrument method and apply appropriate checks. The point 10 is a bench mark having elevation of 66.374 m. (CO4)

Station	Chainage (m)	B. S.	I. S.	F. S.	Remark
1	0	0.597			B. M. = 68.233 m
2	20	2.587		3.132	C. P
3	40		1.565		
4	60		1.911		
5	80		0.376		
6	100	2.244		1.522	C. P
7	120		3.771		
8	140	1.334		1.985	C. P
9	160		0.601		
10	180			2.002	

(c) For the data of leveling operation done previously, determine the loop closure and adjust the calculated values of the levels by applying necessary corrections. Also determine the mean gradient between the points 1 to 10. (CO4)

5. (a) What do you mean by plane surveying ? Enlist various instruments involved in plane table survey. Write down working operations of a plane table survey. (CO5)

(b) Write down different methods of plane table surveying. Explain method of resection after orientation by back ray. (CO5)

(c) Write advantages and disadvantages of plane table surveying. Explain method of resection after orientation by two points. (CO5)

Roll No. ....

TCE–303

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

BUILDING MATERIALS AND CONSTRUCTION TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the major defects in timber with diagram. (CO1)

(b) Explain Lime cycle. Briefly explain how lime influence the properties of cement and brick. (CO1)

(c) Explain the tests to which bricks are generally subjected. (CO1)

2. (a) Explain the procedure of wet and dry process of manufacturing of cement with the help of flowchart. (CO2)

(b) What are admixtures ? Briefly explain its uses. (CO2)

(c) Define aggregates. Enlist the test for aggregates and explain any *two* of them. (CO2)

P. T. O.

3. (a) Explain the orientation and planning of buildings. (CO3)

   (b) Define foundation and briefly explain the function of it. Also write the components of building with the help of a neat sketch. (CO3)

   (c) Define masonry. Enlist the types of bonds used in brick masonry and briefly explain any *two* bonds. (CO3)

4. (a) Explain briefly components of staircase with the help of a neat diagram. (CO4)

   (b) Define window. Briefly explain the types of windows and their use. (CO4)

   (c) Define door. Briefly explain the technical terms used for the components of door with the help of a neat sketch. (CO4)

5. (a) What is construction joint ? Briefly explain the types of joints used in construction. (CO5)

   (b) Write detailed discussion on sound proof construction in terms of characteristics, behavior and acoustic defects. (CO5)

   (c) Write short notes on shuttering, scaffolding and centering. (CO5)

Roll No. ....

TCE–304

B. TECH. (CIVIL ENGG.) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

STRENGTH OF MATERIALS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) A bar of 30 mm diameter is subjected to pull of 80 kN. The measured extension on a gauge length of 250 mm is 0.12 mm and the change in diameter 0.004 mm. Calculate the Poisson's ratio and the value of bulk modulus. (CO1)

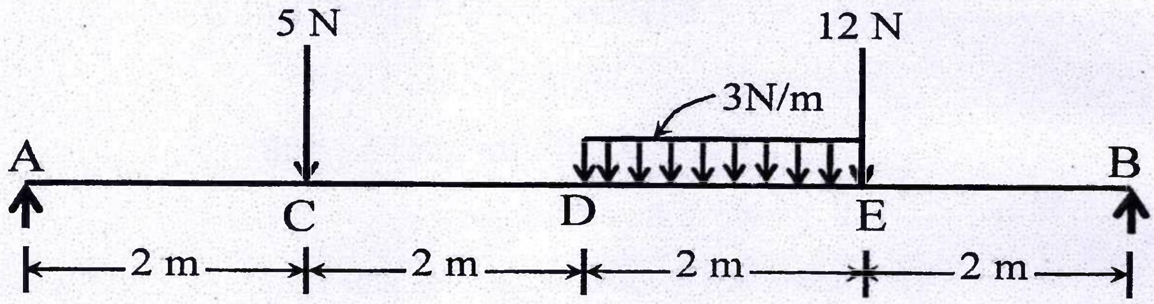
(b) A load of 300 kN is applied on a short concrete column 250 mm × 250 mm. The column is reinforced with 8 bars of 20 mm diameter. If the modulus of elasticity for steel is 15 times that of concrete, find the stresses in concrete and steel. (CO1)

(c) The principal stresses at a point in a bar are 200 N/mm^2 (tensile) and 100 N/mm^2 (compressive). Determine the magnitude and direction of

P. T. O.

resultant stress on a plane inclined at 60° to the axis of major principal stress. Also determine the maximum intensity of shear stress in the material at the point. (CO1)

2. (a) Draw the shear force and bending moment diagrams for the simply supported beam as shown below. Also mark the position of the maximum bending moment and determine its value. (CO2)



(b) Derive an expression for the intensity of shear stress at any point in the cross section of a loaded beam. (CO2)

(c) A 5 m long beam with rectangular section of 15 cm width and 30 cm depth is simply supported at the ends. If it is loaded with a uniformly distributed load ω = 8 kN/m throughout the span and a concentrated load P = 2 kN placed at a distance of 1.5 m from left hand support. Determine the maximum bending stress in the beam. (CO2)

3. (a) A cantilever beam of length 4 meters carries a uniformly distributed load of 2.5 kN per meter for a length of 2 meters from the fixed end and a point load of 1.5 kN at the free end. If the section of the beam is rectangular having 12 cm width and 24 cm depth, find the deflection at the free end. Take E = 210 kN/mm^2. (CO3)

(b) A beam of uniform section is 8 m long and is simply supported at the ends. It carries concentrated loads of 120 kN and 90 kN at distances of 2 m and 4 m respectively from the left end. Calculate the deflection under each load. Take $E = 200$ kN/mm² and $I = 18\times10^8$ mm^4. (CO3)

(c) An aluminium cantilever beam of rectangular section 48 mm wide and 36 mm deep, of length 250 mm carries a uniformly distributed load (w). What will be the maximum value of w if the maximum deflection in the cantilever is not to exceed I mm. Modulus of elasticity for aluminium = 70×10^3 N/mm². (CO3)

4. (a) A solid steel shaft of 12 cm diameter transmits 160 H.P. at 150 r.p.m. Calculate (i) the torque on the shaft (ii) the maximum shear stress (iii) the angle of twist in a length of 80 cm. (CO4)

(b) Show that the hollow circular shaft whose inner diameter is half of the outer diameter has a torsional strength equal to 15/16 of that of a solid shaft of the same outside diameter. (CO4)

(c) Determine the maximum torque that can be applied to a hollow circular steel shaft of 100 mm outside diameter and an 80 mm inside diameter without exceeding a shearing stress of 60 N/mm² or a twist of 0.5 degree per meter. Take modulus of rigidity = 83 kN/mm². (CO4)

5. (a) Derive the expression given by Euler for the crippling load for a column of length (L), modulus of elasticity (E), and moment of inertia (I) when both ends of the column are hinged. (CO5)

(b) A long column of 2 m length is hinged at both ends. Yielding on outer fibres starts when the central deflection is equal to 15 mm. Determine the breadth (b) and depth (d) of the rectangular section with $b/d = 0.4$. Take yield stress = 250 N/mm² and E = 2×10^5 N/mm². (CO5)

(c) A thin cylindrical shell with following dimensions is filled with a liquid at atmospheric pressure : Length = 1.2 m, external diameter = 20 cm, thickness of metal = 8 mm. Find the value of the pressure exerted by the liquid on the walls of the cylinder and the hoop stress induced if an additional volume of 25 cm³ of liquid is pumped into the cylinder. Take E = 2.1×10^5 N/mm² and Poisson's ratio = 0.33. (CO5)

Roll No. ....

TCE–305

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

ENGINEERING MECHANICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

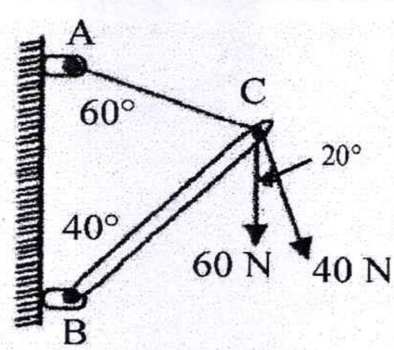
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

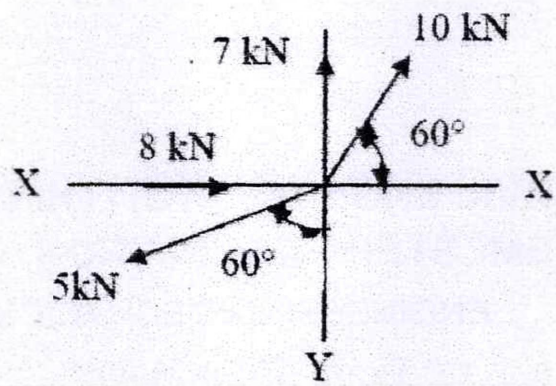
1. (a) State and explain the Principle of Transmissibility of Forces. (CO1)

(b) Two loads are applied as shown in the figure to the end of C of boom BC. Determine the tension in cable AC, knowing that the resultant of three forces exerted at C must be directed along BC. (CO1)



P. T. O.

(c) Find resultant of a force system shown in Figure : (CO1)

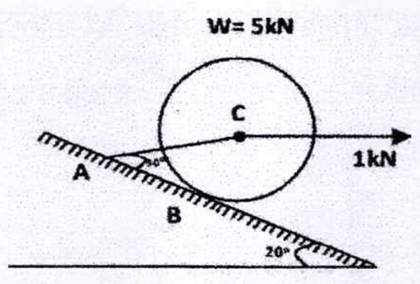


2. (a) Define the following : (CO2)

   (i) Varignon's principle of moments

   (ii) Lami's Theorem (With proof)

(b) A right circular roller of weight 5 kN rests on a smooth inclined plane and is held in position by a chord AC as shown in the figure. Find the tension in the chord and reaction at B , if there is a horizontal force 1 kN acting at C. (CO2)



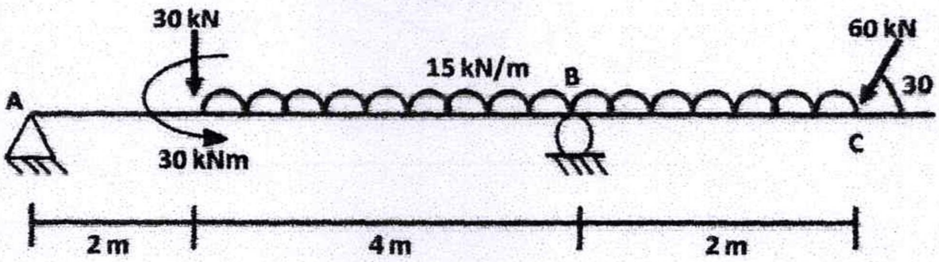
(c) Define the following : (CO2)

   (i) Collinear forces

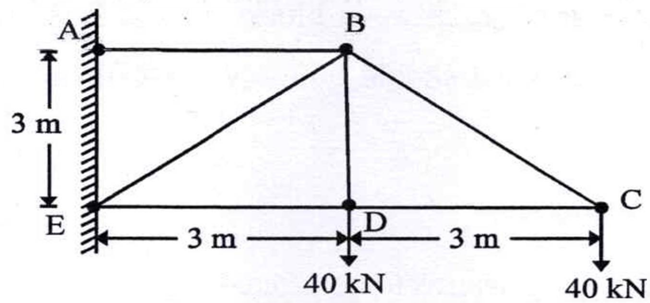
   (ii) Concurrent force system

3. (a) Describe the various types of loading on beams with the help of diagrams. (CO3)

(b) Determine the reactions at support A and B for the beam loaded as shown in figure : (CO3)



(c) Find the forces in all the members of the truss shown in figure. (CO3)



4. (a) Define the following : (CO4)

(i) Radius of gyration

(ii) Polar moment of inertia

(iii) Moment of inertia of an area

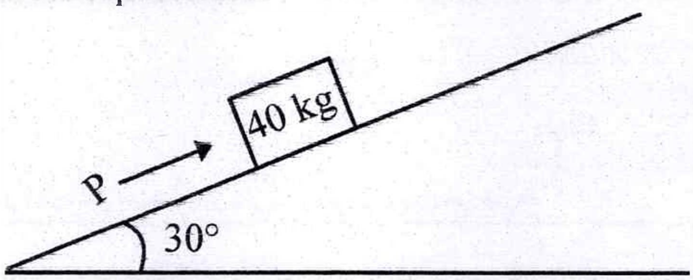
(b) Calculate the centroid of the sector of a circle. (CO4)

(c) Calculate the centroid of the triangle. (CO4)

5. (a) Elaborate the various types of friction. (CO5)

(b) A 40 kg mass is placed on the inclined plane making angle of 30° with horizontal, a shown in figure. A push P is applied parallel to the plane.

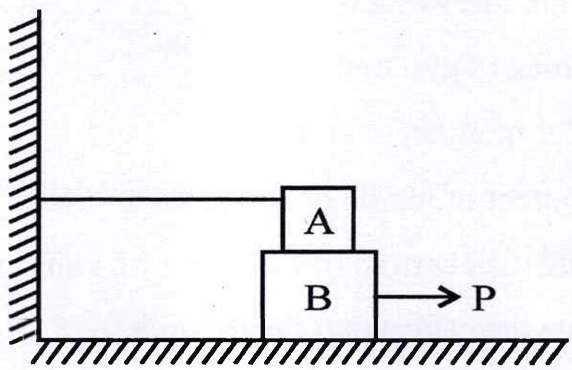
If co- efficient of static friction between the plane and the mass is 0.25, find the maximum and minimum value of P between which the mass will be in the equilibrium. (CO5)



(c) Block A weighing 1000 N rests over block B which weighs 2000 N as shown in figure. Block A is tied to wall with a horizontal string. If the coefficient of friction between blocks A and B is 0.25 and between B and floor is 1/3, what should be the value of P to move the block (B), if : (CO5)

(i) P is horizontal

(ii) P acts at 30° upwards to horizontal



Roll No. ....

TMA–302

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

ENGINEERING MATHEMATICS–III

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Prove that $u(x,y) = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. Find a function $v(x,y)$ such that $f(z) = u + iv$ is analytic. (CO1)

(b) Define the analytic function, and hence to construct an analytic function $f(z)$ using Milne-Thomson method, where the real part of function $f(z)$ is given below : (CO1)

$$u(x,y) = e^{-x}\left[\left(x^2 - y^2\right)\cos y + 2xy \ \sin y\right].$$

P. T. O.

(c) Find the value of the integral $\int_C (x+y)dx + x^2 y\, dy$ along the following curves : (CO1)

(i) $y = x^2$ having (0, 0) and (3, 9) end points,

(ii) $y = 3x$ between the same points.

2. (a) Use Cauchy's integral formula to evaluate the value of a complex integration $\int_C \frac{z}{\left(z^2 - 3z + 2\right)} dz$, where C is the circle $|z-2| = \frac{1}{2}$. (CO2)

(b) Determine whether the function $f(z) = u + iv$, where : (CO1)

$$f(z) = \begin{cases} \frac{x^4(1+i) - y^4(1-i)}{x^3 + y^3}, & z \neq 0 \\ 0 & , \quad z = 0 \end{cases}$$

is analytic or not at $z = 0$.

(c) To evaluate the value of the integral $\oint \frac{e^{zt}}{z^2 - 1} dz$, along the closed curve $C : |z| = 2$. (CO2)

3. (a) Using Newton-Raphson method to compute a root of a non-linear equation $x^3 - x - 1 = 0$ with initial condition $x_0 = 1.5$. (Correct to three decimal places). (CO3)

(b) The velocity (v) of a particle at distance (s) from a point on its path is given by the following table : (CO3)

Distance (s in metres)	Velocity (v in m/sec)
0	47
10	58
20	64
30	65
40	61
50	52
60	38

Calculate the time taken to travel 60 meter by using Simpson's one-third rule.

(c) Perform only six iterations of Bisection method to compute a real root of the equation $x^3 - 5x + 3 = 0$ correct to three decimal places. (CO3)

4. (a) If the probability density function of a continuous random variable X is given by : (CO4)

$$f(x) = \begin{cases} kx^2; & 0 < x < 1 \\ 0\ ; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance, and standard deviation.

(b) The number of airplane landings at an airport in a minute (X) and their probabilities are given by : (CO4)

X_i	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the mean and standard deviation of X.

(c) Following data depicts the statistical values of rainfall and production of wheat in a region for a specified time period : (CO5)

	Mean	Standard Deviation
Production of Wheat (kg/unit area)	10	8
Rainfall (cm)	8	2

Estimate the production of wheat when rainfall is 9 cm if correlation coefficient between production and rainfall is given to be 0.5

5. (a) The following data represent the Carbon dioxide (CO_2) emission from coal-fluid boilers (in units of 1000 ton) over a period of years 1965, and

1977. The independent variable (years) has been standardized yield the following table :

Years (x)	CO_2 emission (y)
0	910
5	680
8	520
9	450
10	370
11	380
12	340

To fit simple linear regression line using the above data. (CO6)

(b) Using least square method to fit a curve $y = ax^2 + bx + c$ for the following data : (CO6)

x	y
1	10
3	5
5	20
10	25
15	22

(c) Using Least square method to fit a curve $y = a + bx$ for the following data : (CO6)

x	y
1	10
3	5
5	20
10	25
15	22